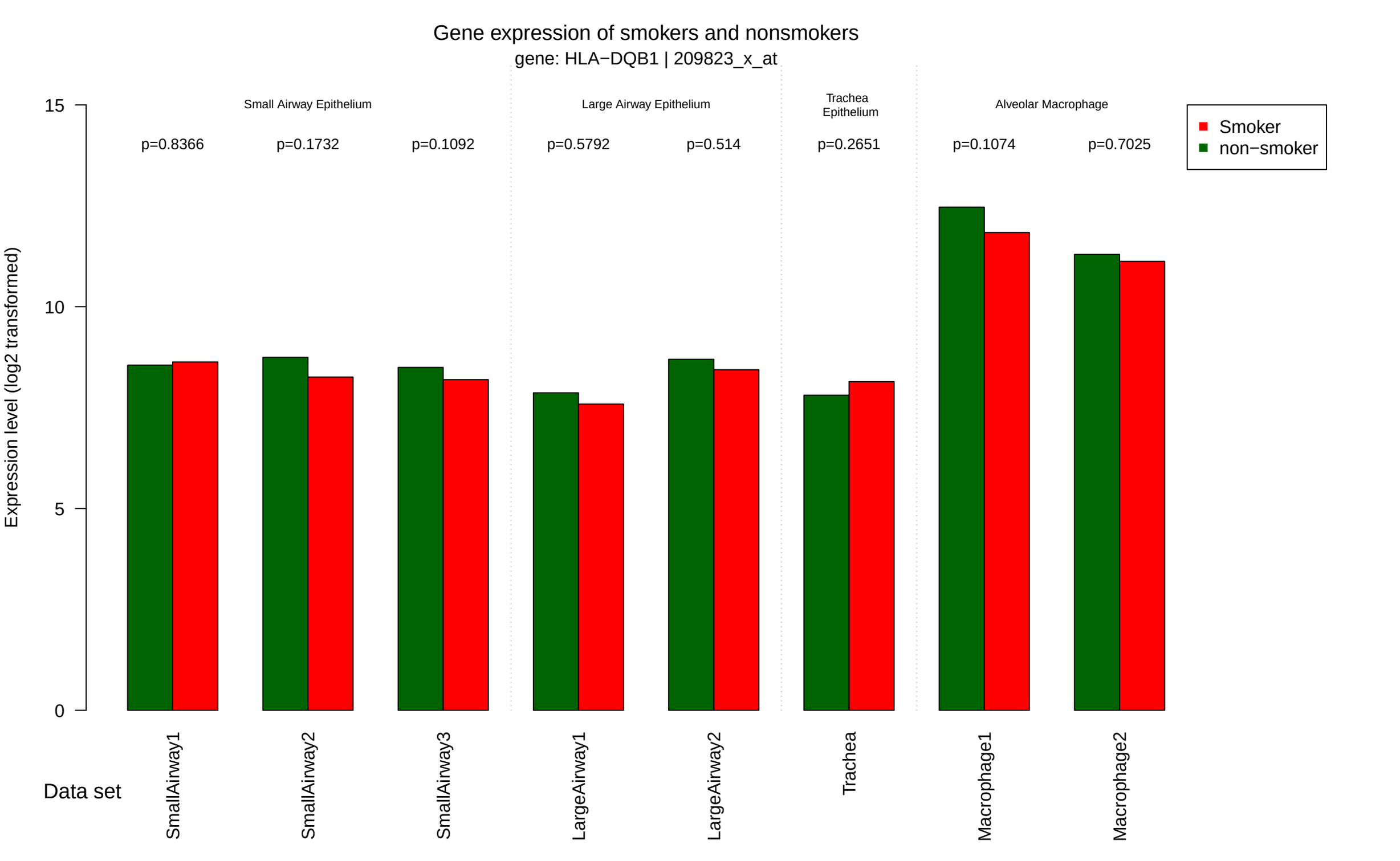
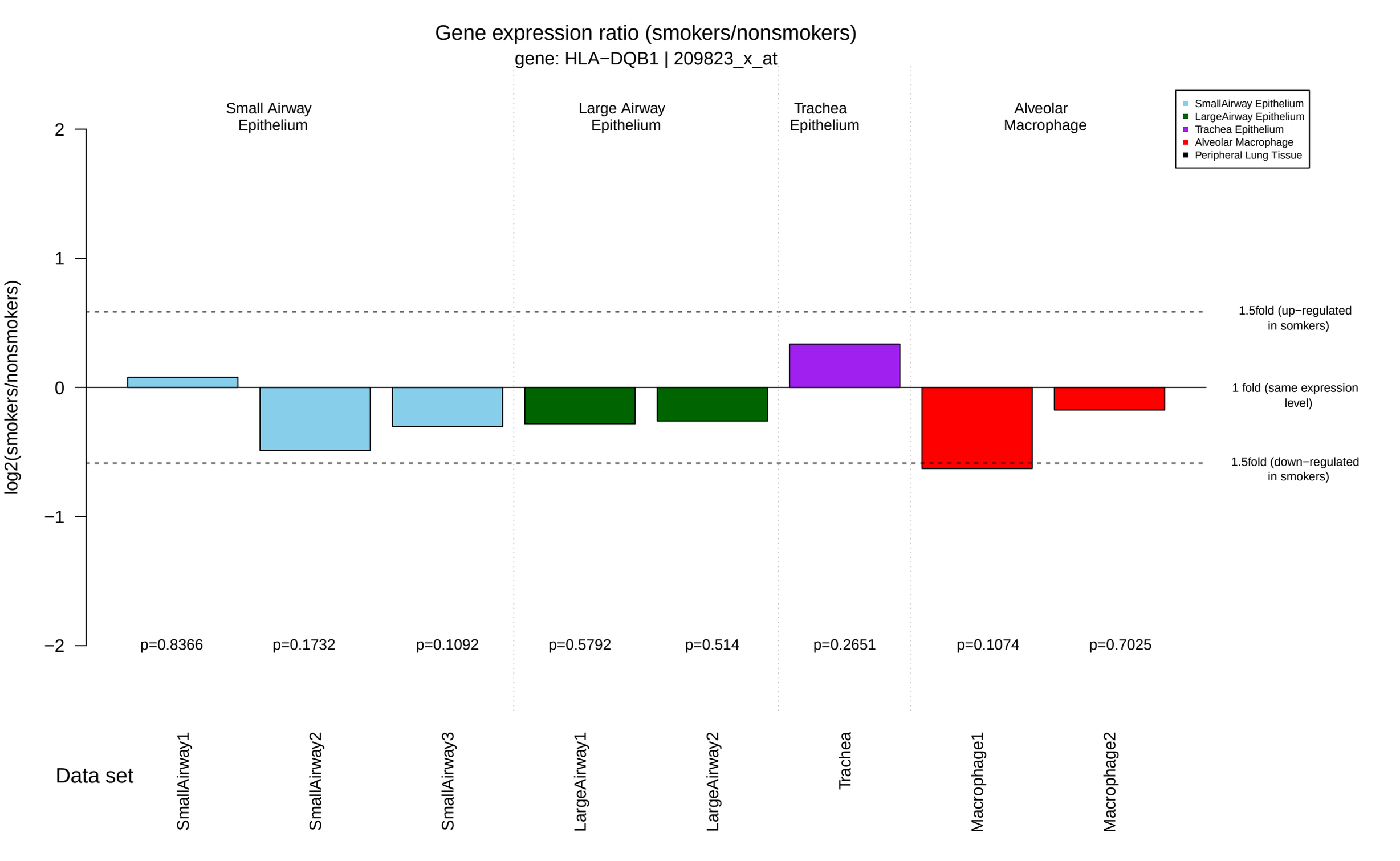
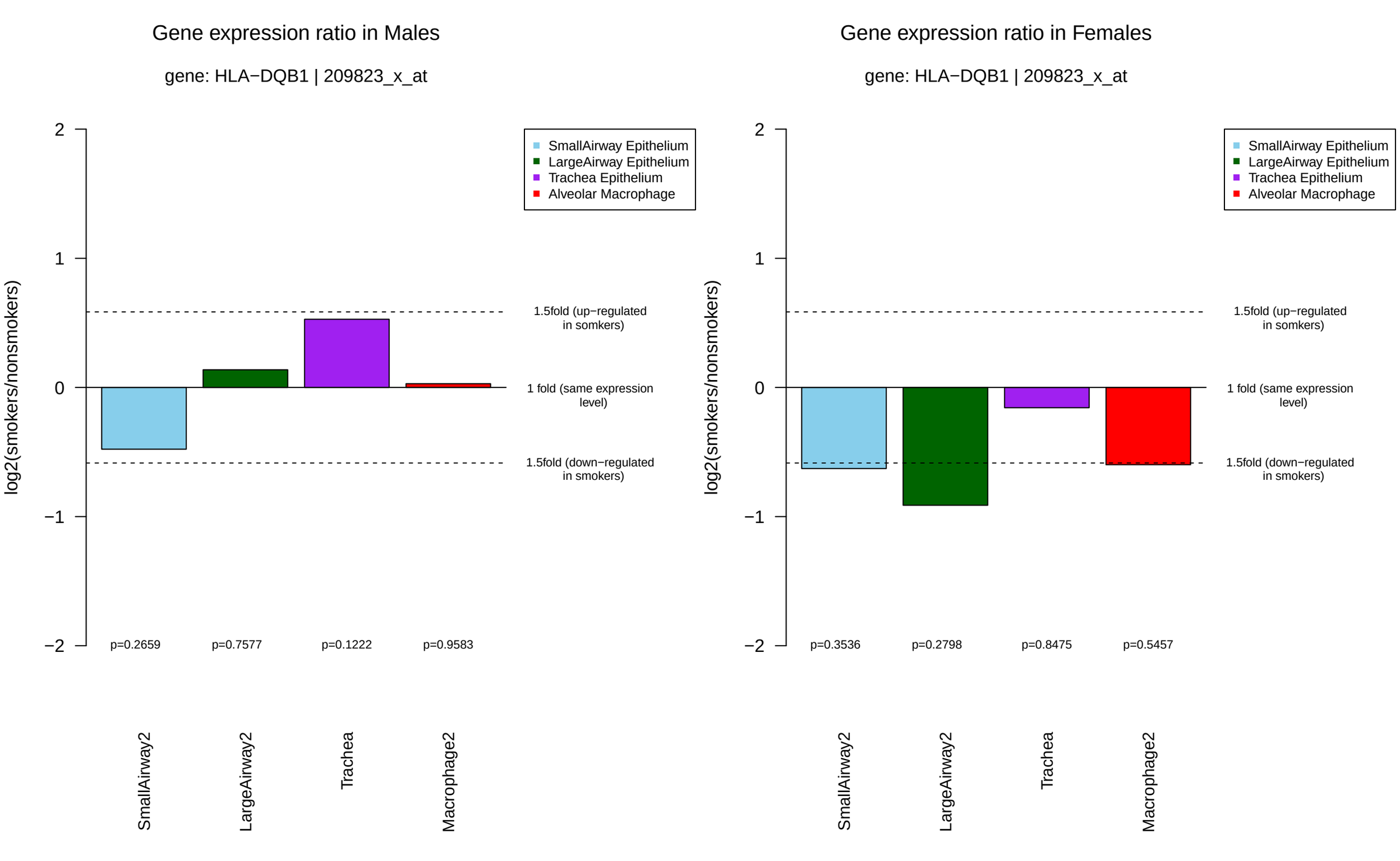
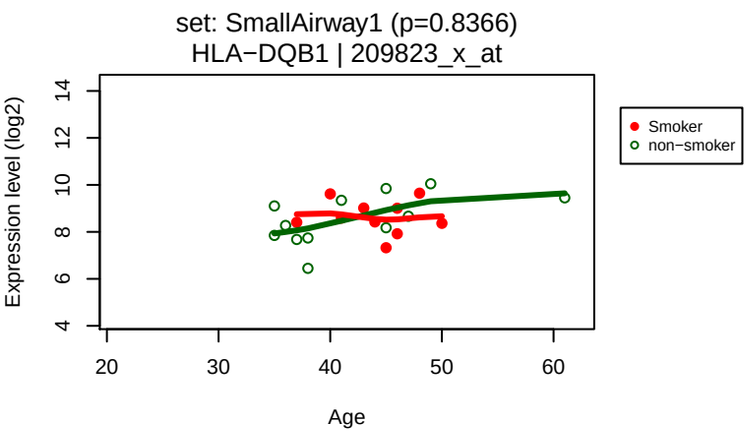
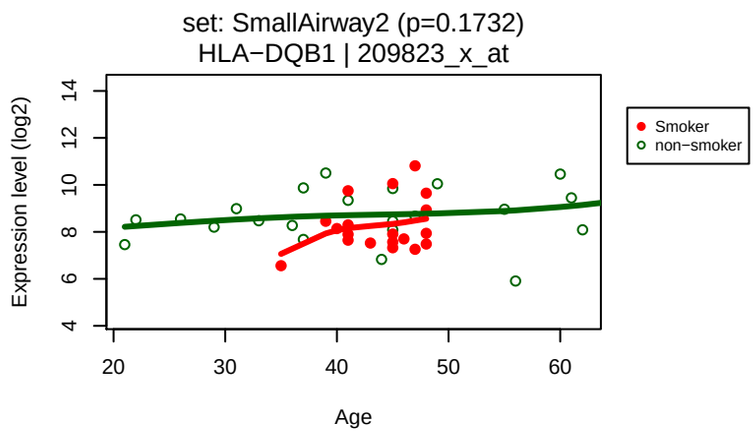
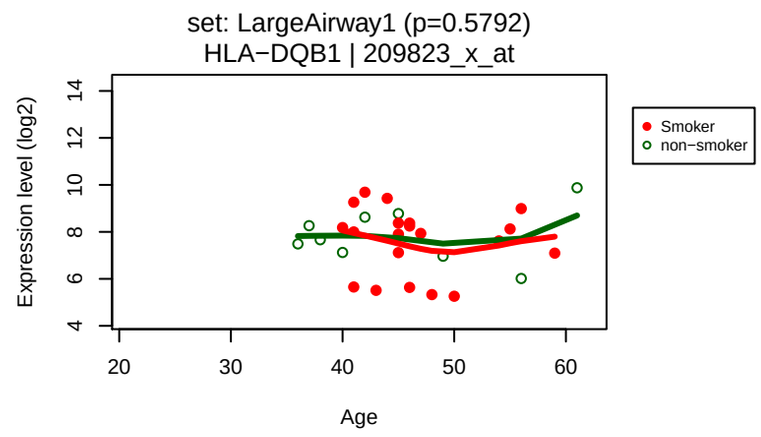

Gene expression of smokers and nonsmokers
gene: HLA−DQB1 | 209823_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8366
p=0.1732
p=0.1092
p=0.5792
p=0.514
p=0.2651
p=0.1074
p=0.7025
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

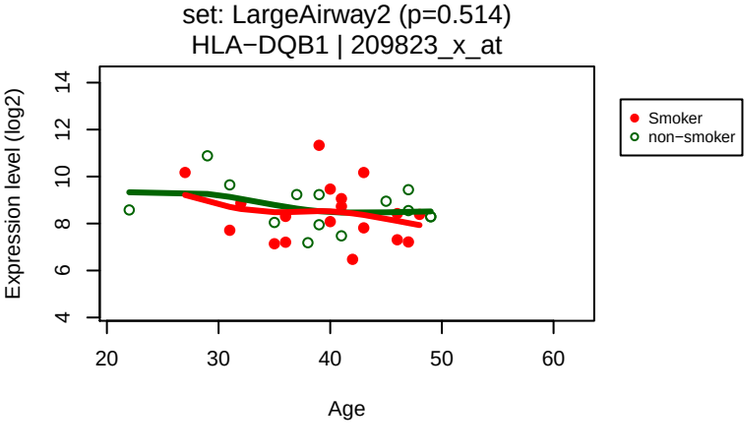

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB1 | 209823_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8366
p=0.1732
p=0.1092
p=0.5792
p=0.514
p=0.2651
p=0.1074
p=0.7025
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

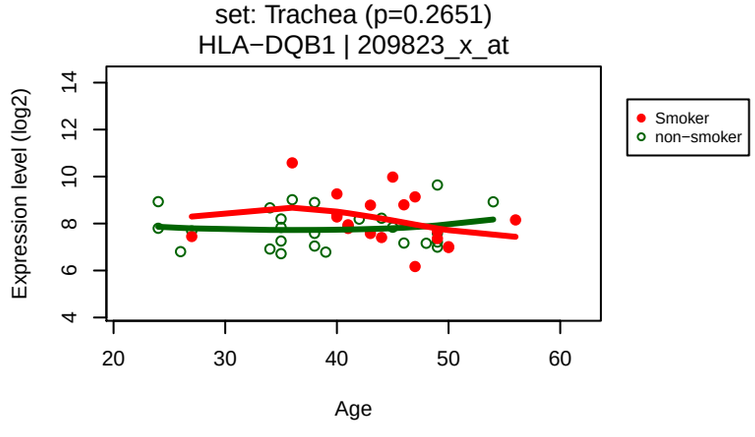

Gene expression ratio in Males
gene: HLA−DQB1 | 209823_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2659
p=0.7577
p=0.1222
p=0.9583
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB1 | 209823_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3536
p=0.2798
p=0.8475
p=0.5457
SmallAirway2
LargeAirway2
Trachea
Macrophage2

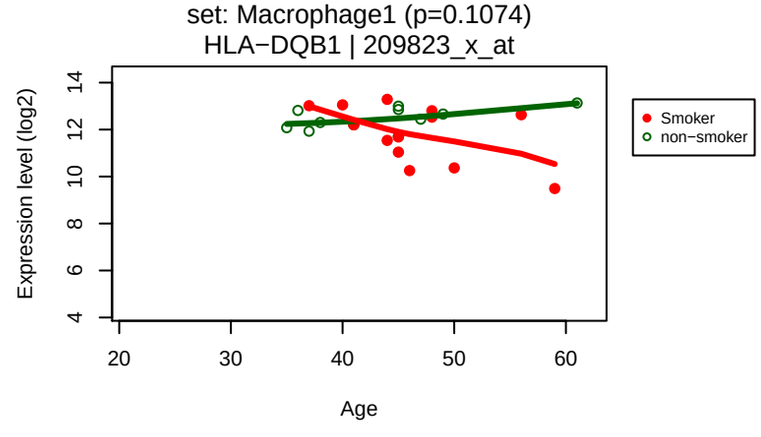

set: SmallAirway1 (p=0.8366)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker

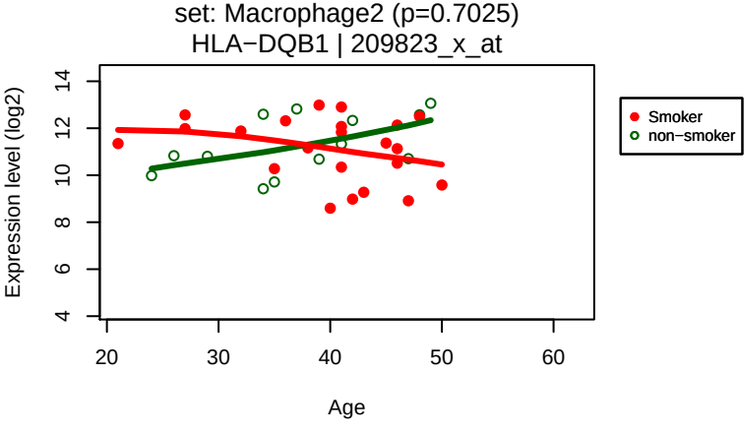

set: SmallAirway2 (p=0.1732)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker

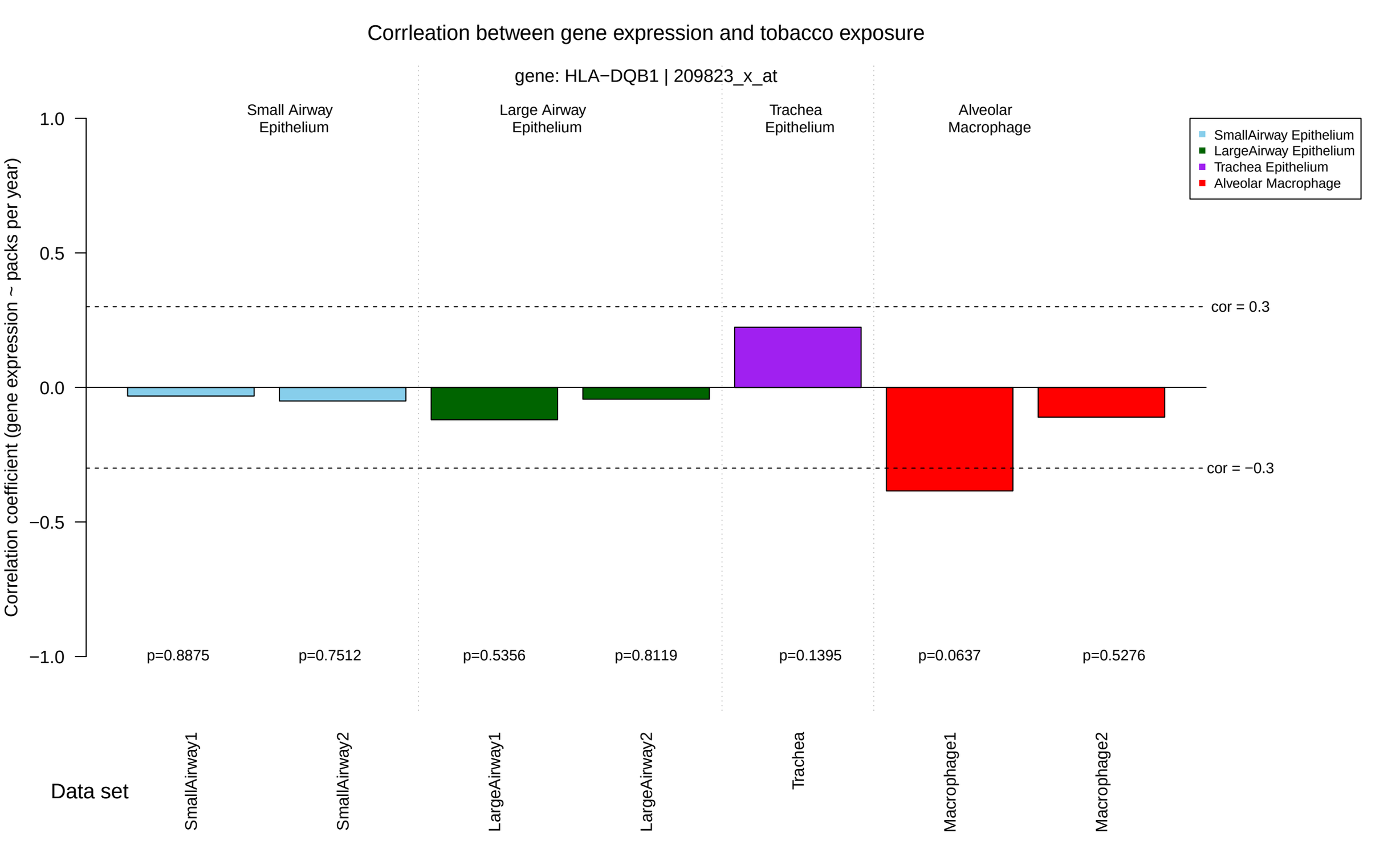

set: LargeAirway1 (p=0.5792)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.514)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2651)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1074)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7025)
HLA−DQB1 | 209823_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB1 | 209823_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8875
p=0.7512
p=0.5356
p=0.8119
p=0.1395
p=0.0637
p=0.5276
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2